AEGIS-X : AI-Powered Disaster Detection and Smart Evacuation Robot

1. **Project Category:** Artificial Intelligence (AI), Robotics, Disaster Management and smart safety system.

2. **Methodology:** the prototype will use an ESP32 microcontroller connected to flame, MQ-2 smoke/gas, tempreture and ultrasonic sensors. When a potential hazard is detected a buzzer warning LEDs, and a display will activate. A route-planning algorithm will identify unsafe corridors and select an alternative exit. The robot will follow predefined paths and avoid obstacles. Wi-Fi may send alerts to a connected device. Testing will be conducted in safe, controlled conditions.

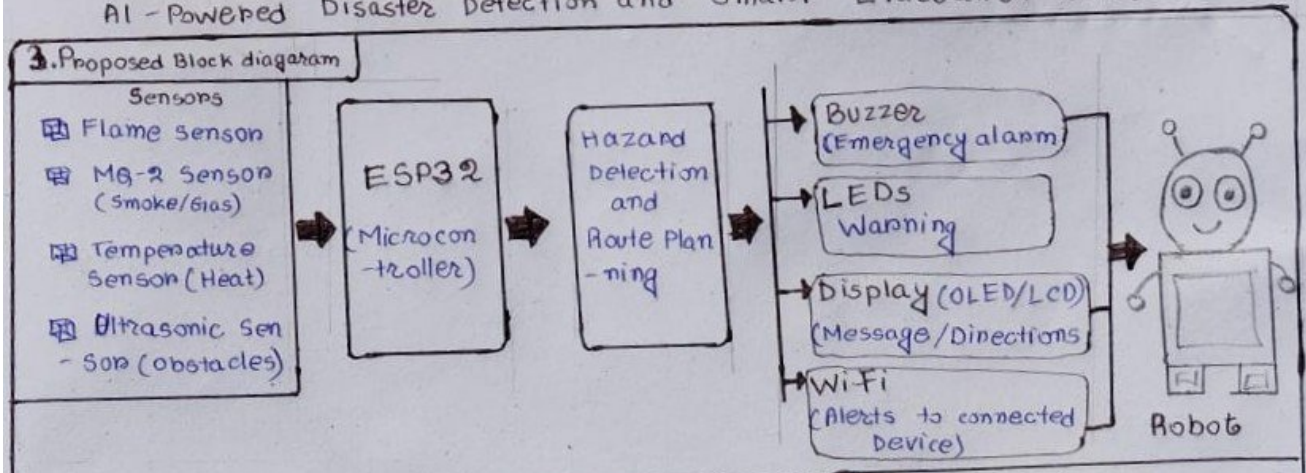
3. **Proposed Block Diagram:** Sensors (Flame, Smoke/Gas, Temperature, Ultrasonic) → ESP32 Microcontroller → Hazard Detection and Route Planning → Robot Movement, Buzzer, LEDs, Display and WiFi Alerts

4. **Prototype:** A miniature building with multiple corridors, exits, sensors, and a mobile robot will demonstrate hazard detection, obstacle avoidance, emergency alerts and alternative evacuation routes.

5. **Proposed Outcome:** The prototype aims to demonstrate faster hazard identification, suitable alternative route selection, and basic smart evacuation. Performance will be evaluated through response time, route-selection accuracy, obstacle avoidance and false alerts. It is an educational prototype, not a certified life-safety device.

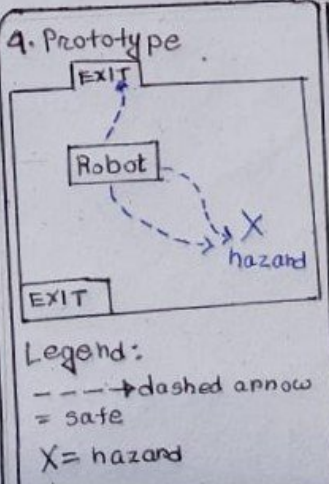
AEGIS-X

AI-Powered Disaster Detection and Smart Evacuation Robot



1. **Project category**
Artificial Intelligence (AI), Robo-tics, Disaster Managements & smart Safety Systems.

2. **Methodology**
1. Collect senson data (flame, MQ-2, temperature, ultrasonic).
2. Detect hazard and activate buzzer, LEDs and display.
3. Use route-planning algorithm to find alternative exit.
4. Follow predefined paths and avoid obstacles.
5. Send WiFi alerts (if available)
6. Test in safe, controlled conditions.



5. **Proposed Outcome**
1. Faster hazard identification
2. Suitable alternative route selection.
3. Basic smart evacuation.
4. Performance will be evalua-ted through:
— Response time
— Route-selection accuracy
— obstacle avoidance
— False alerts.

It is an educational Prototy-pe, not a certified life-safety device.